

U2U-internship-project

Predictive Analytics for Road Accident Prevention

Project Overview

Predictive Analytics for Road Accident Prevention is a Machine Learning project that analyzes historical road accident data to identify accident patterns and predict accident severity.

The system uses factors such as **weather conditions, road surface conditions, light conditions, vehicle type, number of vehicles involved, and number of casualties** to predict the possible severity of an accident.

Project Links

- **GitHub Repository:** U2U-internship-project
- **Source Code:** [View Source Code](#)
- **Dataset:** [View Dataset](#)

Objectives

The main objectives of this project are:

-  Analyze historical road accident data.
-  Clean and preprocess the accident dataset.
-  Perform feature engineering to create useful predictive features.
-  Develop Machine Learning models for accident severity prediction.
-  Evaluate and compare different Machine Learning models.
-  Predict accident severity based on given conditions.
-  Develop an interactive Streamlit application for prediction.

Technologies Used

Programming Language

- Python

Data Analysis

- Pandas

- NumPy

Data Visualization

- Matplotlib
- Seaborn

Machine Learning

- Scikit-learn
- Random Forest
- Decision Tree
- Logistic Regression

Model Management

- Joblib

Web Application

- Streamlit

Development Tools

- Visual Studio Code
- Git
- GitHub

Project Structure

U2U-internship-project/

|

|— README.md

|

|— data/

| |— README.md

| |— raw/

| | |— road_accidents.csv

| |

```
|   └─ processed/
|     └─ processed_accidents.csv
|
|   └─ src/
|     └─ README.md
|     └─ check_data.py
|     └─ data_analysis.py
|     └─ feature_engineering.py
|     └─ train_model.py
|     └─ predict.py
|     └─ app.py
|
|   └─ models/
|     └─ accident_model.pkl
|
|   └─ reports/
|     └─ README.md
|     └─ figures/
|       └─ accident_severity_distribution.png
|       └─ weather_vs_severity.png
|       └─ confusion_matrix.png
|       └─
|     └─ final_report.pdf
|
|   └─ deployment/
|     └─ README.md
|     └─ requirements.txt
```

|
└─ .gitignore

 Project Workflow

Historical Accident Data

|



Data Collection

|



Data Cleaning

|



Exploratory Data Analysis

|



Feature Engineering

|



Data Preprocessing

|



Machine Learning Models

|



Model Evaluation

|



Severity Prediction



Streamlit Web Application

1. Data Collection

Historical road accident data is collected and stored in:

`data/raw/road_accidents.csv`

The dataset contains information related to road accidents, including environmental, road, vehicle, and accident-related factors.

Important features may include:

- Weather conditions
- Road surface conditions
- Light conditions
- Type of vehicle
- Number of vehicles involved
- Number of casualties
- Accident severity

Note: The exact columns depend on the dataset being used.

2. Data Cleaning

The collected dataset is cleaned before Machine Learning.

The cleaning process includes:

- Removing duplicate records.
- Identifying missing values.
- Handling missing categorical values.
- Handling missing numerical values.
- Checking data types.

- Removing unnecessary columns.
- Preparing the dataset for analysis and modeling.

The cleaned dataset is stored in:

data/processed/processed_accidents.csv

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand accident patterns and relationships between different factors.

The analysis focuses on:

Accident Severity

The distribution of different accident severity levels is analyzed.

Weather Conditions

The relationship between weather conditions and accident severity is studied.

Road Conditions

Different road surface conditions are analyzed to identify their relationship with accident severity.

Light Conditions

Daylight and night-time accident patterns are compared.

Vehicle Information

Vehicle types and their relationship with accident severity are analyzed.

Accident Characteristics

The number of vehicles involved and casualties are also studied.

Generated graphs are stored in:

reports/figures/

4. Feature Engineering

Feature engineering is used to transform raw data into useful features for Machine Learning.

Examples include:

- Extracting year from date.

- Extracting month from date.
- Extracting day of the week.
- Extracting accident hour.
- Creating weekend indicators.
- Encoding categorical variables.
- Selecting relevant predictive features.

These features help the Machine Learning model identify patterns in accident severity.

5. Machine Learning

The project uses classification algorithms to predict accident severity.

The following models can be evaluated:

1. Logistic Regression

A baseline classification algorithm used to establish an initial performance level.

2. Decision Tree

A tree-based model that makes predictions using a series of decision rules.

3. Random Forest

An ensemble Machine Learning algorithm that combines multiple decision trees to improve prediction performance.

The best-performing model is selected based on evaluation results.

6. Model Evaluation

The trained models are evaluated using classification metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Example result format:

Model	Accuracy
Logistic Regression	Actual Result
Decision Tree	Actual Result
Random Forest	Actual Result

Note: The actual values should be added after training the models. Results should not be manually estimated.

7. Accident Severity Prediction

After training, the selected Machine Learning model can predict accident severity based on input conditions.

Example Input

Weather: Rainy

Road Condition: Wet

Light Condition: Night

Vehicle Type: Motorcycle

Vehicles Involved: 2

Casualties: 1

Example Output

Predicted Accident Severity: High

The prediction depends on the trained model and the actual dataset.

8. Streamlit Web Application

The project includes an interactive web application built using **Streamlit**.

The application allows users to enter accident-related conditions and receive a predicted accident severity.

Application Features

- Weather condition selection.
- Road condition selection.

- Light condition selection.
 - Vehicle type selection.
 - Number of vehicles input.
 - Number of casualties input.
 - Accident severity prediction.
-

9. How to Run the Project

Step 1: Clone the Repository

```
git clone <[(https://github.com/shivani2004r-cell/U2U-internship-project)]>
```

Move into the project directory:

```
cd U2U-internship-project
```

Step 2: Create a Virtual Environment

```
python -m venv venv
```

Windows PowerShell

```
.\venv\Scripts\Activate.ps1
```

If activation is successful, the terminal will show:

```
(venv)
```

Step 3: Install Dependencies

```
pip install -r deployment/requirements.txt
```

Step 4: Check the Dataset

Go to the source folder:

```
cd src
```

Run:

```
python check_data.py
```

This checks:

- Dataset columns
 - Dataset size
 - Data types
 - Missing values
 - Sample records
-

Step 5: Perform Data Analysis

Run:

```
python data_analysis.py
```

This performs exploratory data analysis and generates graphs.

Step 6: Perform Feature Engineering

Run:

```
python feature_engineering.py
```

The processed dataset will be generated in:

```
data/processed/processed_accidents.csv
```

Step 7: Train the Machine Learning Model

Run:

```
python train_model.py
```

The trained model will be saved in:

```
models/accident_model.pkl
```

Step 8: Test Prediction

Run:

```
python predict.py
```

The program will display the predicted accident severity.

Step 9: Run Streamlit Application

From the src folder:

```
streamlit run app.py
```

The Streamlit application will open in your browser.

Dataset

The project uses a historical road accident dataset for analysis and Machine Learning.

The dataset should be placed at:

```
data/raw/road_accidents.csv
```

The target variable is:

```
Accident_severity
```

The dataset may contain multiple accident severity classes depending on the source dataset.

Expected Output

The project produces:

Data Analysis

Accident severity distribution

Weather vs accident severity

Road condition vs accident severity

Light condition vs accident severity

Machine Learning

Model accuracy

Precision

Recall

F1-score

Confusion matrix

Prediction

Predicted Accident Severity

Web Application

Interactive Streamlit prediction interface

Key Benefits

- Helps identify accident severity patterns.
 - Uses historical data for predictive analysis.
 - Identifies important accident-related factors.
 - Supports data-driven accident analysis.
 - Provides an interactive prediction interface.
 - Demonstrates practical application of Machine Learning.
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Future Enhancements

The project can be improved by adding:

-  Accident hotspot detection.
 -  Geographic accident visualization.
 -  Real-time weather data.
 -  Real-time traffic information.
 -  Mobile application.
 -  Real-time risk alerts.
 -  Cloud deployment.
 -  Interactive dashboards.
 -  GPS-based accident risk prediction.
 -  More advanced Machine Learning models.
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Limitations

- Prediction quality depends on the quality and coverage of the historical dataset.

- Historical patterns may not represent every current road situation.
- The model does not guarantee that an accident will occur.
- Predictions should be treated as analytical estimates.
- Real-time traffic, weather, and geographic information may not be included in the initial version.

Project Information

Project: U2U Internship Project

Title: Predictive Analytics for Road Accident Prevention

Domain: Data Science & Machine Learning

Programming Language: Python

Framework: Streamlit

Repository: GitHub

Conclusion

The **Predictive Analytics for Road Accident Prevention** project demonstrates how historical road accident data can be combined with Data Science and Machine Learning to analyze accident patterns and predict accident severity.

By combining **data analysis, feature engineering, Machine Learning, visualization, and an interactive Streamlit application**, the project provides a practical demonstration of predictive analytics for road accident severity classification.

Project Workflow Summary

Collect Data



Clean Data



Analyze Data



Engineer Features



Train ML Models



Evaluate Models



Select Best Model



Predict Accident Severity



Streamlit Application

Acknowledgement

This project was developed as part of the **U2U Internship Project** to demonstrate practical skills in Data Science, Machine Learning, and predictive analytics.